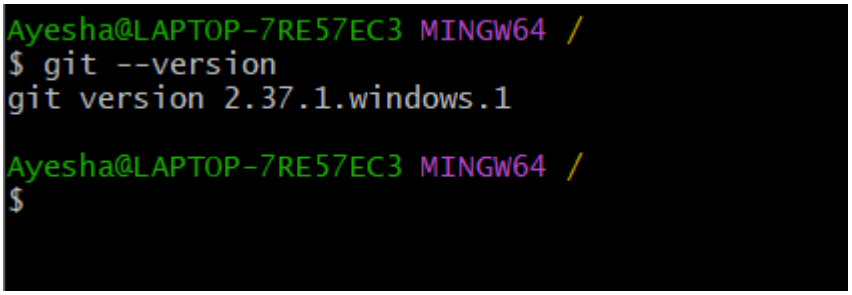
1. Install Git

sudo apt update
sudo apt install git -y
git --version

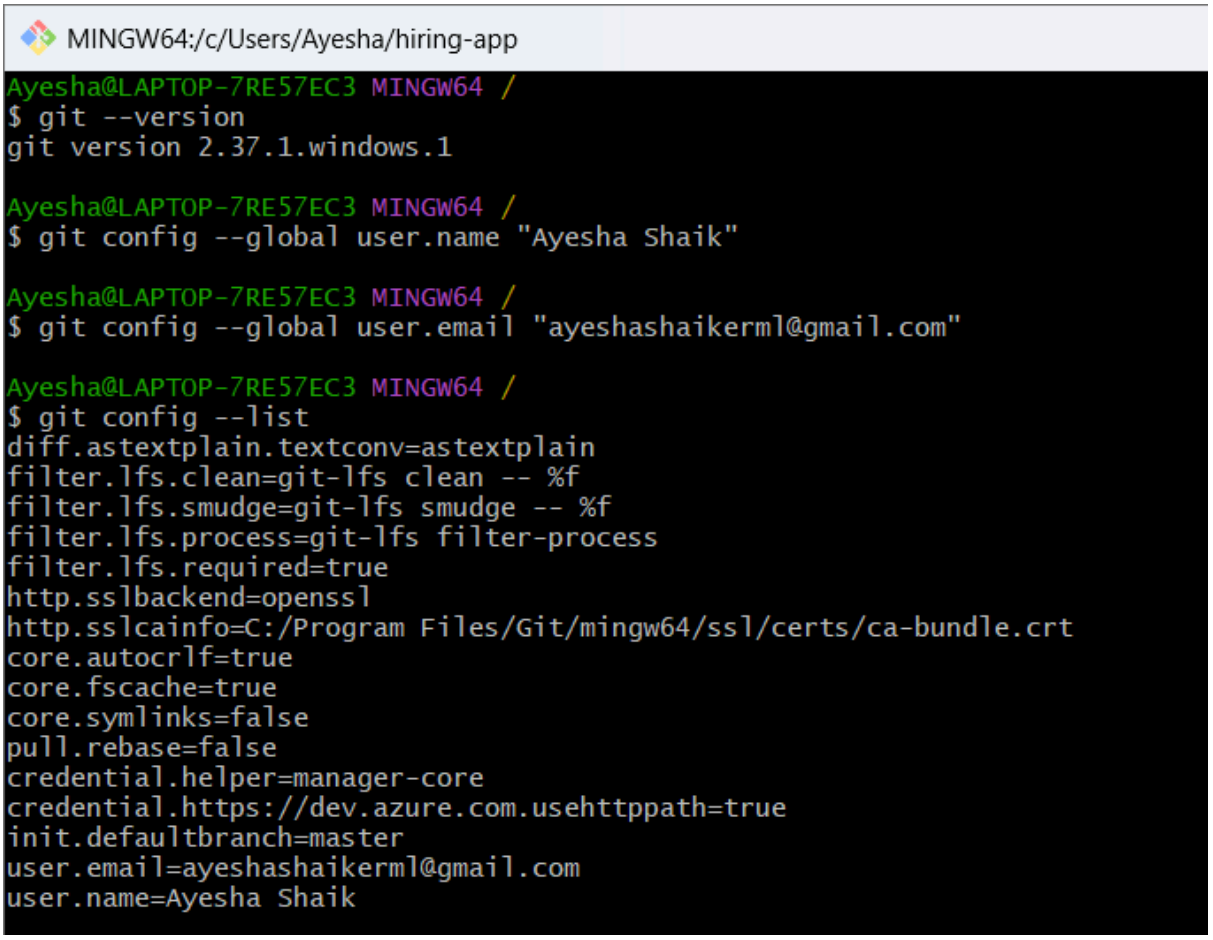
I used `sudo apt update` to update the package information. Then I used `sudo apt install git -y` to install Git. Finally, I used `git --version` to verify that Git was installed successfully.



2. Configure Git

git config --global user.name "Ayesha Shaik"
git config --global user.email "ayeshashaikerml@gmail.com"
git config --list

I used `git config --global user.name` to configure my Git username. I used `git config --global user.email` to configure my email address. These details are associated with the commits I create. I used `git config --list` to verify my configuration.



3. Create a project directory

mkdir hiring-app
cd hiring-app
pwd
ls
ls -la

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~ (master)
$ mkdir hiring-app

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~ (master)
$ cd hiring-app

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ pwd
/c/Users/Ayesha/hiring-app

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ ls

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ ls -la
total 16
drwxr-xr-x 1 Ayesha 197121 0 Oct 22 14:46 ./
drwxr-xr-x 1 Ayesha 197121 0 Oct 22 14:46 ../

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `mkdir hiring-app` to create my project directory. Then I used `cd hiring-app` to enter the directory. I used `pwd` to check my current location and `ls` to see the files. I used `ls -la` to also display hidden files.

4. Initialize Git

git init
git status

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git init
Initialized empty Git repository in C:/Users/Ayesha/hiring-app/.git/

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `git init` to initialize Git in my project directory. This created a hidden `.git` directory, which stores Git's repository information. Then I used `git status` to check the current status of my repository.

5. Create files

touch A B C
git status

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ touch A B C

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        A
        B
        C

nothing added to commit but untracked files present (use "git add" to track)

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `touch A B C` to create three files. Then I used `git status` to check the repository. Git showed these files as untracked because I had not yet added them to Git's staging area.

6. Add files to staging

git add A
git add A B C
git add .
git status

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git add A

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git add A B C

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git add .

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   A
        new file:   B
        new file:   C


Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `git add` to move files from the working directory to the staging area. I can add one file, multiple files, or all files using `git add .`. Then I used `git status` to verify which files were staged.

7. Commit the changes

git commit -m "Initial commit"

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git commit -m "Initial commit"
[master (root-commit) cbe1c82] Initial commit
 3 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 A
 create mode 100644 B
 create mode 100644 C

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `git commit` to permanently save the staged changes in the Git repository. I provided `"Initial commit"` as the commit message so I could understand what changes were made.

8. Check commit history

git log

git log --oneline
git log -4

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git log
commit cbe1c82d000c0289d0a70d79c06a0745d6d212cc (HEAD -> master)
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 14:51:49 2026 +0530

    Initial commit

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git log --oneline
cbe1c82 (HEAD -> master) Initial commit

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git log -4
commit cbe1c82d000c0289d0a70d79c06a0745d6d212cc (HEAD -> master)
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 14:51:49 2026 +0530

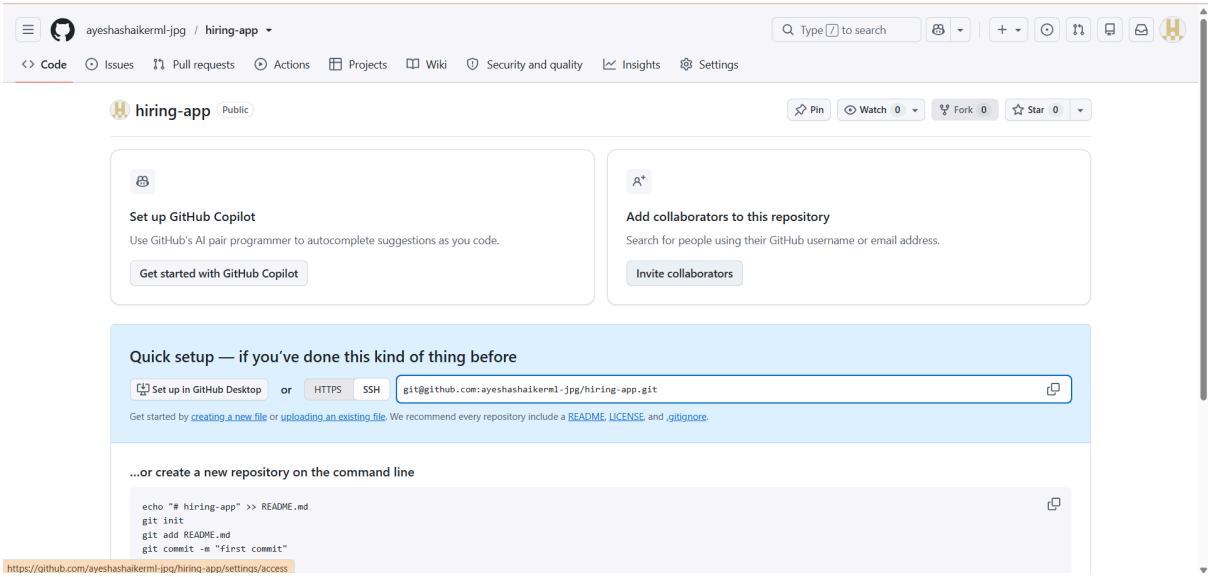
    Initial commit

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I used `git log` to view the complete commit history. I used `git log --oneline` to see the commits in a short format. I used `git log -4` to display the latest four commits.

9. Create GitHub repository and connect it

After creating the repository on GitHub, I used:



```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git remote add origin git@github.com:ayeshashaikerml-jpg/hiring-app.git

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git remote -v
origin  git@github.com:ayeshashaikerml-jpg/hiring-app.git (fetch)
origin  git@github.com:ayeshashaikerml-jpg/hiring-app.git (push)

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$
```

I created a repository named `hiring-app` on GitHub. Then I used `git remote add origin` to connect my local Git repository with the GitHub repository. I used `git remote -v` to verify the remote URL.

10. Rename branch to main

git branch
git branch --show-current
git branch -M main

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git branch
* master

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (master)
$ git branch -M main

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git branch
* main

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$
```

I used `git branch` to check my current branch. I used `git branch --show-current` to display the current branch name. Then I used `git branch -M main` to rename my branch to `main`.

11. Push code to GitHub

git push -u origin main

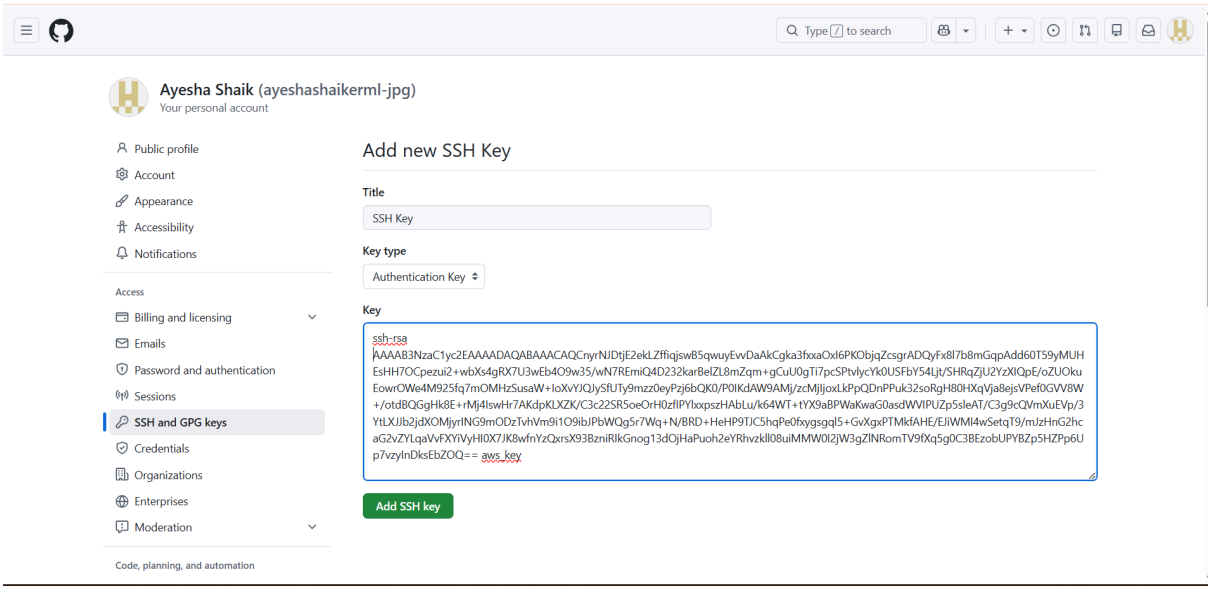
```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git push -u origin main
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ ls -la ~/.ssh
total 44
drwxr-xr-x 1 Ayesha 197121    0 Oct 22 09:41 ./
drwxr-xr-x 1 Ayesha 197121    0 Oct 22 15:01 ../
-rw-r--r-- 1 Ayesha 197121 3369 Oct 12 15:10 id_rsa
-rw-r--r-- 1 Ayesha 197121  733 Oct 12 15:10 id_rsa.pub
-rw-r--r-- 1 Ayesha 197121 4596 Oct 22 15:02 known_hosts
-rw-r--r-- 1 Ayesha 197121 4332 Oct 22 09:41 known_hosts.old

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQCnyrNJDtjE2ekLZffiqjswB5qwuyEvvDaAkCgka3fxxaOxI6PKObjqZcsgrADQyFx
8l7b8mGqpAdd6OT59yMUHEsHH7OCpezui2+wbXs4gRX7U3wEb4O9w35/wN7REmiQ4D232karBelZL8mZqm+gCuUOgTi7pcSPtvlycYk
OUSFbY54Ljt/SHRqZjU2YzXIQpE/oZUOkuEowrOWe4M925fq7mOMHzSusaW+IoXvYJQJySfUTy9mzzOeyPzj6bQKO/POIKdAW9AMj/z
cMjIjoxLkPpQDnPPuk32soRgH8OHXqVja8ejsVPefOGVV8W+/otdBQGgHk8E+rMj4IswHr7AKdpKLXZK/C3c22SR5oeOrHOzfIPYlxx
pszHAbLu/k64WT+tYX9aBPWaKwaGOasdWVIPUZp5sleAT/C3g9cQVmXuEVp/3YtLXJJb2jdXOMjyrING9mODzTvhVm9i1O9ibJPbWQg
5r7Wq+N/BRD+HeHP9TJC5hqPeOfxygsgql5+GvXgxPTMkfAHE/EJiWMI4wSetqT9/mJzHnG2hcaG2vZYLqaVvFXYiVyHIOX7JK8wfnY
zQxrsX93BzniRIkGnog13dOjHaPuoh2eYRhvzkll08uiMMWOI2jW3gZlNRomTV9fXq5gOC3BEzobUPYBZp5HZPp6Up7vzyInDksEbZO
Q== aws_key

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$
```




```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 221 bytes | 73.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:ayeshashaikerml-jpg/hiring-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

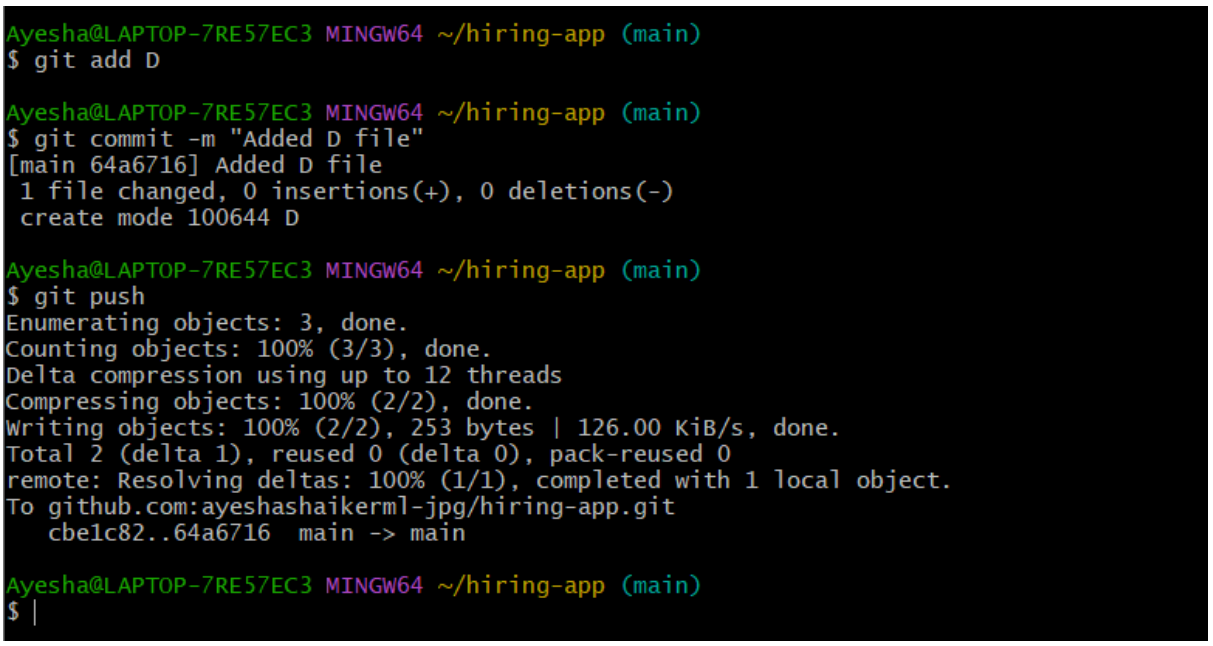
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git push
Everything up-to-date

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$
```

I used `git push -u origin main` to upload my local commits to the GitHub repository. The `origin` represents my remote GitHub repository, and `main` represents the branch I want to push.
I used `git push` to upload my subsequent local commits to GitHub.

12. Make another change

touch D
git status
git add D
git commit -m "Added D file"
git push



I created a new file called `D`. I used `git status` to check the change. Then I used `git add D` to stage the file, `git commit` to save the change, and `git push` to upload the new commit to GitHub.

13. Pull changes from GitHub

git pull

I used `git pull` to download the latest changes from the remote GitHub repository and integrate those changes into my current local branch.

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git pull
Already up to date.

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 900 bytes | 47.00 KiB/s, done.
From github.com:ayeshashaikerml-jpg/hiring-app
   64a6716..ba3bd1c  main       -> origin/main
Updating 64a6716..ba3bd1c
Fast-forward
 E | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 E

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$
```

I used `git pull` to download the latest changes from the remote GitHub repository and integrate those changes into my current local branch.

14. Fetch changes

git fetch
git fetch origin

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git fetch

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git fetch origin
```

> I used `git fetch` to download information about changes from the remote repository without automatically merging those changes into my current branch.

Easy difference:

- `git fetch` → downloads changes but does **not** merge.
- `git pull` → downloads changes and **integrates** them.

15. Clone a repository

git clone git@github.com:YOUR_USERNAME/hiring-app.git
cd hiring-app

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ git clone git@github.com:ayeshashaikerml-jpg/hiring-app.git
Cloning into 'hiring-app'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 8 (delta 2), reused 4 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), done.
Resolving deltas: 100% (2/2), done.

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app (main)
$ cd hiring-app

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$
```

I used `git clone` when I wanted to get an existing GitHub repository onto my local machine. It downloads the repository, including its files and Git history.

16. Unstage a file

git reset F

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ touch F

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git add A

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ ls
A  B  C  D  E  F

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git add F

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git reset F

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        F

nothing added to commit but untracked files present (use "git add" to track)

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$
```

I used `git reset F` when I wanted to remove file `F` from the staging area without deleting the file from my working directory.

17. Revert a commit

git log --oneline
git revert <commit-id>

```
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 15:09:33 2026 +0530

    Added new

commit 64a67160974ccf8dda7065ce81909857e88c1752
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 15:07:45 2026 +0530

    Added D file

commit cbe1c82d000c0289d0a70d79c06a0745d6d212cc
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 14:51:49 2026 +0530

    Initial commit

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git revert ba3d1
fatal: bad revision 'ba3d1'

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git revert ba3bd1c8f0837815ba18e66a18a63476284746b4
[main ef99b31] Revert "Added new"
 1 file changed, 1 deletion(-)
 delete mode 100644 E

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$ git log
commit ef99b319248dad7972a85f09ec35c63248a051e3 (HEAD -> main)
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 15:20:54 2026 +0530

    Revert "Added new"

    This reverts commit ba3bd1c8f0837815ba18e66a18a63476284746b4.

commit ba3bd1c8f0837815ba18e66a18a63476284746b4 (origin/main, origin/HEAD)
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 15:09:33 2026 +0530

    Added new

commit 64a67160974ccf8dda7065ce81909857e88c1752
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 15:07:45 2026 +0530

    Added D file

commit cbe1c82d000c0289d0a70d79c06a0745d6d212cc
Author: Ayesha Shaik <ayeshashaikerml@gmail.com>
Date:   Thu Oct 22 14:51:49 2026 +0530

    Initial commit

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~/hiring-app/hiring-app (main)
$
```

I first used `git log --oneline` to find the commit ID. Then I used `git revert <commit-id>` to undo the changes introduced by that commit. Git creates a **new commit** to reverse the previous changes.